Route Optimisation Engine: Idea & Research Directions

Hi everyone! This doc explains the idea behind our project and what I have found so far.

The idea is simple. A company tells our engine where it needs to go (the stops), which vehicles it has, what is a must, and what is most important. The engine then finds the best routes. It looks at traffic, road conditions and the size of each vehicle. We build one core engine and adjust it for each partner, because a parcel company and an e-scooter company like Voi need very different things.

Nothing here is final. I did some research to give us a starting point, but I want your ideas. If you think something is wrong, or you find a better way, please say it. That is the most useful thing you can do right now.

The idea, and where I'd start

Below are my first guesses for the big questions. Use them as a starting point, not as answers.

Open question	My starting guess	Why it's worth testing
Road engine	Valhalla (MIT)	Truck height, weight, axle load and hazmat set per request; time dependent speeds; bike and scooter profiles
Solver	OR-Tools default, PyVRP quality mode, VROOM baseline	OR-Tools covers the most constraints for free; PyVRP is 0.2 to 0.8% from best known solutions; VROOM runs in days
Priorities	Three level score: hard violations, then unserved priority, then weighted soft cost	Keeps "must" separate from "nice to have" so weights can't break necessities
Traffic	Travel time matrices in 15 minute buckets; historical averages first, LightGBM second	No free street level probe data in Sweden, so start simple and log data from day one
Data	OSM + NVDB + Trafikverket API + Stockholm open data + SMHI	Free, mostly CC0, covers restrictions, incidents, road condition, weather
First vertical	Voi battery swap and rebalancing	Stockholm HQ, one of only two permitted operators since 1 July 2026, problem testable offline
December deliverable	Working engine, a Voi style demo on real Stockholm roads, a benchmark report	Pitch material for a pilot in Q1 2027

Things I don't think we should do (but tell me if you disagree): build our own solver from zero, use neural network solvers, or build truck routing for Scania or Volvo, because they already build their own.

What I want from you

Choose one track below. Read about its questions, try things out, and bring what you find to your 1:1 meeting with me in the week of 20 Oct. Ideas outside these tracks are welcome too.

Track	Questions to dig into	Bring back
Algorithm	Is OR-Tools, PyVRP or something else the better core for our constraints? Would you model priorities differently from the three level score? Can traffic go inside the search, not just after it? Is there a hybrid or learned approach worth trying?	A small comparison on Solomon instances, or a counter proposal
Data	Where could we get street level speeds for Stockholm? How good are OSM and NVDB for truck restrictions? What does Trafikverket really cover? Which datasets did I miss?	A data map, ideally a first Valhalla or OSRM build on Sweden
Interface	How should a non expert set priorities and weights? What does a dispatcher need to see? How should drivers, or vehicles, receive routes? What do Routific or OptimoRoute get right and wrong?	Sketches or a clickable mock
Partners (anyone)	Which vertical has the clearest pain and data we could actually get? Is Voi right, or is there a better first partner? What would they measure us on?	A one pager on one vertical

The problem, formalised

A “necessity” is something the route must do (a hard rule). A “priority” is something we want, but can give up if needed (a soft rule). Here is one way to turn what a user says into rules for the solver:

What the user says	Model term	How we encode it (OR-Tools)
“Must visit this stop”	Required stop	Left out of every disjunction (docs)
“This one is urgent”	Priority class	Drop penalty per class, spaced about 10x apart
“Deliver between 10 and 12”	Hard time window	`CumulVar(i).SetRange(a, b)`
“Ideally before 11”	Soft time window	`SetCumulVarSoftUpperBound`, coefficient scaled by class
“Truck is 4 m tall, 18 t”	Vehicle profile	Separate matrix per profile from Valhalla truck costing; forbidden roads cost infinity
“Only the van with a cooler”	Skills	`SetAllowedVehiclesForIndex`
“Swap any 5 of these 12 scooters”	Choose k of n	`AddDisjunction(idxs, penalty, max_cardinality=k)`
“Pick up here, drop there”	Pickup and delivery	`AddPickupAndDelivery`
“Fewer vans, shorter days, balanced routes”	Weighted objectives	Arc cost weights, span cost, fixed vehicle cost

The objective (The Math AI suggested)

$$\min_{x} \; (H(x),\; P(x),\; S(x)) \quad \text{in strict (lexicographic) order}$$

$$P(x) = \sum_{i \in U(x)} \pi_{c(i)} \qquad S(x) = \sum_{k} w_k \frac{f_k(x)}{s_k}$$

- **H** counts hard violations. The engine only returns plans with H = 0; otherwise it reports which constraint made the problem infeasible.
- **P** sums the penalty of every unserved stop U, by class c (critical, high, normal, low). Classes are spaced about 10x so one high stop outweighs all normal ones.
- **S** is the soft cost: drive time, distance, class weighted lateness, overtime, vehicles used, route imbalance. Each term f is divided by a scale s taken from a reference solve, so weights mean the same thing across partners.

Solvers: what's out there

Starting guess: OR-Tools for constraint coverage, PyVRP for solution quality. Untested.

Solver	Licence	Language	Strong at	Weak at
OR-Tools 9.15	Apache 2.0	C++, Python	Widest constraint set; warm start and route locks for replanning	Quality drops above about 500 stops (14% gap at 1,000 stops after 1 min, vendor benchmark)
PyVRP 0.14	MIT	Python on C++	0.2 to 0.8% from best known; prizes, required stops, reloads, multi depot	No soft windows, no breaks, no time dependent travel
VROOM 1.15	BSD 2	C++, JSON API	100 stops in about 0.4 s; priority 0 to 100; skills; shipments	No soft windows, no time dependence, fixed cost model
Timefold 2.5	Apache 2.0 core, paid tiers	Java	Any constraint in code, native score levels, live changes	Key scaling features are Enterprise only; Python binding archived
jsprit 2.0	Apache 2.0	Java	Time dependent travel	Small community
LKH-3, Hexaly, Gurobi, Nextmv	Academic only or commercial	Various	Strong quality	Licences block partner use
Neural solvers (POMO, RouteFinder)	Mostly MIT	PyTorch	Research	Gap jumps from 3.7% to 34% when capacity shifts (paper); not production ready

Road network, vehicle size and traffic

Starting guess: Valhalla, for per request truck dimensions and time dependent speeds. OSRM if it's too slow; benchmark both.

Engine	Licence	Vehicle size	Time dependent	Self hosted matrix
Valhalla 3.9	MIT	Per request: height, width, length, weight, axle load, axle count, hazmat; bicycle, scooter, pedestrian	Yes: weekly profiles in 5 min buckets plus live traffic file	Yes, slower (2017 test: 1,000x1,000 in about 111 s; reworked since, must re-measure)
OSRM 26.9	BSD 2	Baked in at build time, one dataset per vehicle class, no hazmat	No; swap speeds by re-running customise	Yes, fastest (10k x 10k about 8 s reported)
GraphHopper 11	Apache 2.0	Per request custom models, truck and cargo bike files	No traffic in open source	No, commercial only
OpenRouteService 9.10	GPL 3	driving-hgv profile	No	Yes, slow for HGV

Data sources

Most of the data we need is free. The missing piece is traffic speed on small city streets in Stockholm.

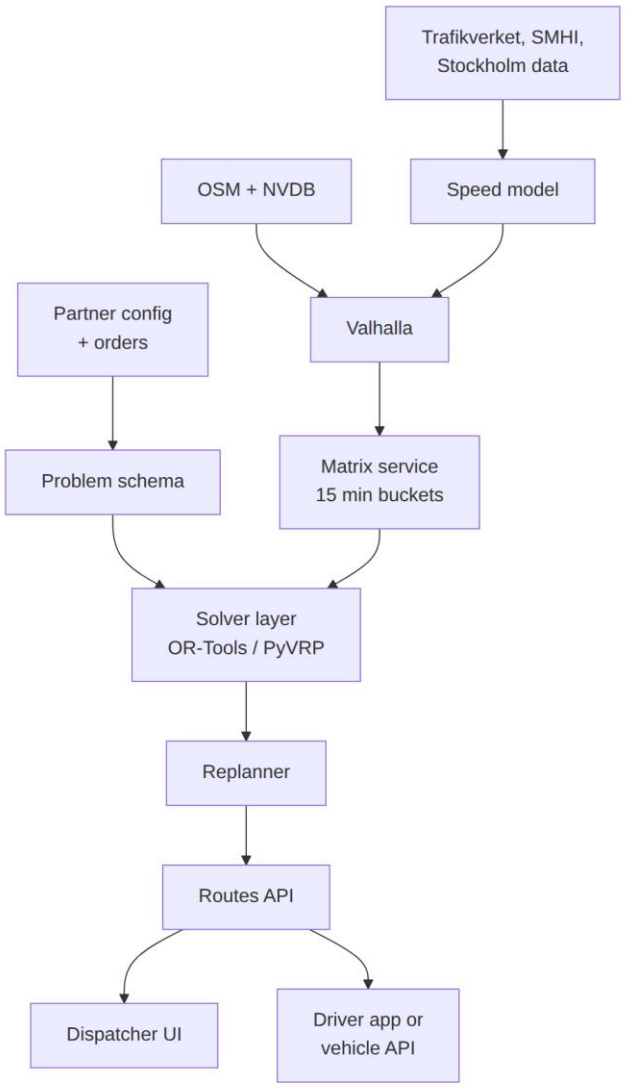
Source	What it gives us	Access	Use
Trafikverket Open API	Situation (roadworks, accidents), TrafficFlow, TravelTimeRoute, RoadCondition, weather stations, ferries	Free key, CC0, no hard quota	Live layer, speed history
NVDB in the Open API and Lastkajen	Bearing class (BK1 64 t to BK4 74 t), height limits, weight, width and length limits, banned turns, speed limits, environmental zones	Free, CC0, GeoPackage	Truck restrictions, better than OSM tags
Stockholm open street data	Loading zones, truck parking, height limits, street works, e-scooter parking, traffic counts	Free key; moving to OGC API Features in 2026	Last 50 m rules
Stockholm parking rules API	Truck parking, street cleaning bans	Free key	Stop feasibility
SMHI forecasts	SNOW1gv1 weather forecast (the old pmp3g API shut 31 Mar 2026)	No key	Winter speed factors
Trafiklab KoDa	Historical SL bus positions since Dec 2020	Free key	Congestion proxy on city streets
TomTom free tier	Traffic flow tiles, routing with traffic	Free up to monthly limits	Ground truth spot checks
OSM Sweden	Base road graph	ODbL	Graph, joined with NVDB

Benchmarks and test data

Dataset	What it tests	Licence
Solomon and Gehring & Homberger	Time windows, 25 to 1,000 stops, best known solutions	Free
CVRPLIB X and XL	Capacity, 100 to 10,000 stops	Free
Li & Lim	Pickup and delivery	Free
Amazon Last Mile 2021	9,184 real routes with real travel times	CC BY-NC, research only
LaDe	10.7M real deliveries with GPS	Apache 2.0
SUMO	Simulates Stockholm traffic from OSM to test plans under congestion	EPL 2.0

A possible architecture

This is only a sketch. The main idea: every part uses one shared format to describe the problem. That way the same result can go to a driver's phone app or straight to a self driving vehicle.



Rough shape to December

This is a rough plan, not a promise. The first three weeks are for exploring. We decide the later steps together, based on what you find.

Phase	Dates	Focus	Outcome
0 · Explore and propose	28 Sep to 18 Oct (3 wks)	Each track researches its questions, tries a tool or two, and brings ideas	Proposals at the kickoff; we agree the approach together in the 1:1s
1 · First working engine	19 Oct to 8 Nov (3 wks)	Whatever stack we agreed, end to end on Stockholm roads	A real Stockholm instance solved with priorities and drawn on a map
2 · Traffic and replanning	9 Nov to 29 Nov (3 wks)	Time of day travel times, replanning when things change	Evidence that traffic awareness actually improves plans
3 · Demo and interface	30 Nov to 20 Dec (3 wks)	A partner style scenario, dispatcher UI, benchmark report	A demo and report we can take to a partner
Next · Pilot	Jan to Mar 2027	Real partner data	Partner signs off on a shadow run

Partners and market

Many companies already sell general route planning, so I think we should start with one type of business first. “Priority” means very different things in each one:

Vertical	Priority in practice	Fit for us now
Micromobility (Voi)	Rides lost if a scooter stays flat, weighted by predicted demand per zone and hour; cost of breaking parking rules	Best
Parcel (PostNord, DHL)	SLA tier, time window, cost of a failed first attempt	Saturated and cheap
Food (Wolt, Foodora)	Freshness, batching, one courier at a time dispatch	Core to their business, real time
Heavy trucks (Scania, Volvo)	Bearing class, height, EU driving time rules (9 h a day, 45 min break after 4.5 h)	They build in house
Home care	Carer continuity, skills, care decision windows	Health data under GDPR, Caire already there

Open questions

- Vertical first, or stay general longer?
- First partner: Voi, or something with easier data?
- Where do we get street level speeds?
- Server: the MLOps project’s box or our own?
- What did this doc miss?

Sources

Solvers

- OR-Tools releases, penalties and dropping visits, time windows

- PyVRP benchmarks, PyVRP paper
- VROOM benchmarks, VROOM API
- Timefold pricing, Timefold score levels
- Hexaly CVRPTW benchmark (vendor)
- Neural solver generalisation study

Routing engines

- Valhalla API reference, Valhalla speeds, Valhalla matrix
- OSRM, OSRM traffic
- GraphHopper custom models, OpenRouteService
- Geofabrik Sweden, TomTom pricing, Google Routes billing

Swedish data

- Trafikverket Open API, NVDB in the Open API, Lastkajen, bearing classes
- Stockholm open street data, Miljözon class 3, e-scooter rules
- SMHI forecast API, Trafiklab KoDa

Benchmarks and simulation

- Solomon, Gehring & Homberger, CVRPLIB, Li & Lim
- Amazon Last Mile, LaDe, SUMO

Modelling research

- Ichoua et al., time dependent VRP, Marler & Arora, weighted sums
- Google Maps ETA with graph nets, STID
- ReEvo, ARS natural language to VRP

Market

- Routific, OptimoRoute, Nextmv, Google Route Optimization, HERE Tour Planning
- Caire, Einride Saga
- SVT on Stockholm scooter permits, Voi Q2 2025
- Lee, Lee & Park 2024, swap and rebalance, Koca & Sharafi 2026